

## LESSON 6-2 • SESSION 2 • PRACTICE • FORM A

## Dividing with Mixed Numbers

Learning target – I can divide mixed numbers by writing them as improper fractions first. • 6.NS.A.1

## REMEMBER

- Write each mixed number as an **improper fraction**:  
 $2\frac{3}{4} = (2 \times 4 + 3)/4 = 11/4$ .
- Then **multiply by the reciprocal** of the divisor.
- Simplify. Write an improper answer as a **mixed number**.

## WORKED EXAMPLE

A ribbon is  $6\frac{3}{4}$  meters long. Each bow uses  $2\frac{1}{4}$  meters. How many bows can be made?

**Step 1** Convert:  $6\frac{3}{4} = 27/4$  and  $2\frac{1}{4} = 9/4$ .

**Step 2** Multiply by the reciprocal:  $27/4 \times 4/9 = 108/36$ .

**Step 3** Simplify:  $108/36 = 3$ . Check:  $3 \times 2\frac{1}{4} = 6\frac{3}{4}$ .

**Answer 3 bows**

**WORDS** mixed number (*número mixto*) • improper fraction (*fracción impropia*) • simplify (*simplificar*)

## PRACTICE

show your work • label units

1. Divide. Write each quotient in simplest form.

- a.  $5\frac{1}{2} \div 1\frac{1}{4} =$  \_\_\_\_\_      b.  $3\frac{3}{8} \div \frac{3}{4} =$  \_\_\_\_\_      c.  $4\frac{1}{2} \div 1\frac{1}{2} =$  \_\_\_\_\_
- d.  $2\frac{1}{2} \div \frac{1}{3} =$  \_\_\_\_\_      e.  $1\frac{3}{4} \div 3\frac{1}{2} =$  \_\_\_\_\_      f.  $2\frac{2}{3} \div 4 =$  \_\_\_\_\_

2. **Garden Box.** A builder has a  $4\frac{1}{2}$ -foot board and  $7\frac{1}{2}$  pounds of soil.

- a. He cuts the board into  $\frac{3}{4}$ -foot pieces. How many pieces? \_\_\_\_\_ pieces
- b. Each bag holds  $2\frac{1}{2}$  pounds of soil. How many bags can he fill? \_\_\_\_\_ bags

Show your work.

3. **Paint Coats.** A painter has  $3\frac{3}{4}$  liters of paint. Each coat uses  $1\frac{1}{4}$  liters. How many coats can she paint?

**Hint**  $3\frac{3}{4} = 15/4$  and  $1\frac{1}{4} = 5/4$ .

Show your work.

**Answer** \_\_\_\_\_ coats

4. **Yarn Scarves.** Bea has  $11\frac{1}{4}$  yards of yarn. Each scarf needs  $2\frac{1}{4}$  yards. How many scarves can she knit?

**Hint** Write both as fourths first.

Show your work.

**Answer** \_\_\_\_\_ scarves

5. **Sugar Scoops.** A jar holds  $4\frac{1}{2}$  cups of sugar. Tomás says it fills  $2\frac{1}{4}$  half-cup scoops. Justify or correct him.

**Hint** More scoops or fewer scoops than cups?

Show your work.

**Answer** \_\_\_\_\_ scoops

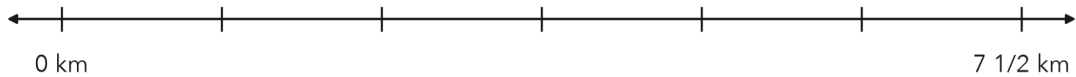
6. **Door Trim.** A shop has  $16\frac{1}{2}$  ft of trim. A door uses  $4\frac{1}{2}$  ft. The rest is cut into  $1\frac{1}{2}$ -ft pieces.

**Hint** Subtract first, then divide by  $\frac{3}{2}$ .

- a. How much trim is left? \_\_\_\_\_ ft
- b. How many pieces? \_\_\_\_\_ pieces

Show your work.

- 7. Trail Markers.** A trail is  $7\frac{1}{2}$  km long. Markers are placed every  $1\frac{1}{4}$  km, starting at the  $1\frac{1}{4}$  km point. Ticks are every  $1\frac{1}{4}$  km.



- a. How many  $1\frac{1}{4}$ -km spacings fit along the trail? \_\_\_\_\_ spacings
- b. How many markers are placed, counting the one at the end? \_\_\_\_\_ markers
- c. Why are the two counts the same? \_\_\_\_\_
- 8. Punch Pitchers.** A jug holds  $9\frac{1}{3}$  cups of punch. Each pitcher holds  $2\frac{1}{3}$  cups. How many pitchers can be filled? Check by multiplying.

**Hint** Check:  $4 \times 2\frac{1}{3}$  should equal  $9\frac{1}{3}$ .

Show your work.

**Answer** \_\_\_\_\_ pitchers

FIND THE ERROR

explain the mistake, then fix it

- 9. Omar's Work.** Omar solved  $4\frac{1}{4} \div \frac{1}{2}$ . He changed  $4\frac{1}{4}$  into  $\frac{5}{4}$ , then wrote  $\frac{5}{4} \times \frac{2}{1}$ . Explain his mistake. Then find the correct quotient.

TEST PRACTICE

choose the best answer

- 10.** Written as an improper fraction,  $2\frac{3}{4}$  is –
- A**  $\frac{11}{4}$                       **B**  $\frac{6}{4}$                       **C**  $\frac{9}{4}$                       **D**  $\frac{23}{4}$
- 11.** Before you divide mixed numbers, you must –
- A** flip both                      **B** make improper fractions
- C** divide wholes only                      **D** add the wholes

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

|          |          |          |           |          |           |          |              |          |          |
|----------|----------|----------|-----------|----------|-----------|----------|--------------|----------|----------|
| <b>3</b> | <b>5</b> | <b>7</b> | <b>12</b> | <b>9</b> | <b>11</b> | <b>4</b> | <b>2 1/4</b> | <b>6</b> | <b>8</b> |
| <b>F</b> | <b>L</b> | <b>X</b> | <b>P</b>  | <b>I</b> | <b>Z</b>  | <b>D</b> | <b>Q</b>     | <b>E</b> | <b>P</b> |

|    |    |    |     |     |     |    |
|----|----|----|-----|-----|-----|----|
| #3 | #4 | #5 | #6a | #6b | #7a | #8 |
|    |    |    |     |     |     |    |

CHECK YOURSELF      ☐ I can do this on my own      ☐ I need a little help      ☐ I have a question about #\_\_\_\_